

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2020

ESE 3107

(Solar Thermal Engineering)

Time: 1 Hour 30 Minutes.

Full Marks: 120

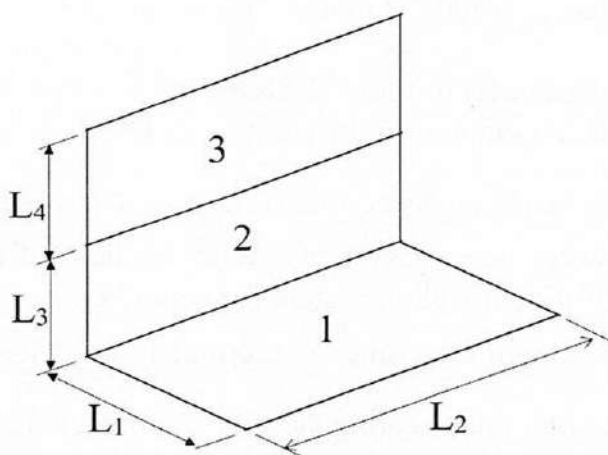
N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). Define the following terms: 09
(i) Sunshine hour, (ii) Clearness index, and (iii) Hour angle
- 1(b). Explain the working principle of pyranometer with schematic diagram. 10
- 1(c). What is solar time corresponding to 11 AM on march 8 at Khulna city (longitude 89.54 °E)? 11
- 2(a). How to calculate the solar radiation on a tilted surface? Explain. 08
- 2(b). What are the features of a blackbody? Show that the emissive power of a blackbody is π times of radiation intensity. 12
- 2(c). What is view factor? Deduce the expression for view factor F_{1-3} as shown in the figure. 10



- 3(a). What is meant by radiosity? Draw the radiation network for a three zone enclosure with a reradiating surface. 10
- 3(b). Explain the construction details of a typical flat plate solar collector with schematic diagram. 10
- 3(c). Explain the working principle of a evacuated tube solar collector. What are its advantages? 10

SECTION – B

- 4(a). Define the following terms: 06
 (i) Concentration ratio, (ii) Critical radiation level, and (iii) Collector efficiency factor.
- 4(b). For a flat plate solar collector, write a short note on ‘Glazing Material’. 08
- 4(c). For a flat plate collector having the following characteristics and ignoring the bond resistance, calculate (i) fin efficiency, (ii) collector efficiency factor, (iii) useful energy, and (iv) collector efficiency: 16
 Collector area = 4 m^2
 Flow rate = 0.06 kg/s
 Overall heat loss coefficient = $6.9 \text{ W/m}^2 \text{ } ^\circ\text{C}$
 Tube spacing = 120 mm
 Tube outside diameter = 15 mm
 Tube inside diameter = 13.5 mm
 Plate thickness = 0.4 mm
 Plate material thermal conductivity = $385 \text{ W/m}^\circ\text{C}$
 Heat transfer coefficient inside the tubes = $320 \text{ W/m}^2 \text{ } ^\circ\text{C}$
 $(\tau\alpha) = 0.8$
 The global solar radiation for 1 hour = 2.88 MJ/m^2
 Collector operates at a temperature difference = 5°C
- 5(a). What is solar tracking? Distinguish between three axis and two axis solar tracking system. 10
- 5(b). Classify the solar water heating systems. With the help of neat sketch, explain the working principle of thermosiphon system for water heating. 12
- 5(c). How does a storage system work in solar thermal system? Explain. 08
- 6(a). What does mean by solar pool-heating system? With the help of a neat sketch, explain the working principle of a direct solar drying system. 12
- 6(b). Explain the central receiver system to provide high temperature for thermal power generation. 08
- 6(c). Solar still has water and glass temperature equal to 60°C and 50°C , respectively. The constants C and n are determined experimentally and found to be $C = 0.04$ and $n = 0.35$. If the convective heat transfer coefficient from water surface to glass is $2.8 \text{ W/m}^2\text{K}$, estimate the heat transfer rate per square meter of still to produce distillate. 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2020

EE 3113

(Power System Engineering)

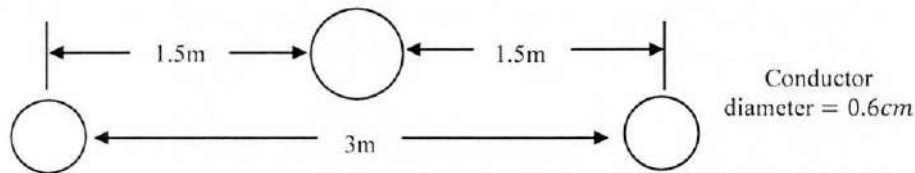
Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any Two questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

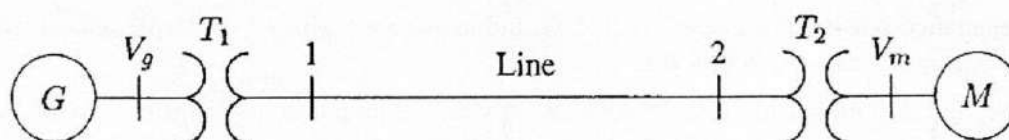
- 1(a). What are the advantages of using ACSR conductors? Determine the inductance of a two wire transmission line due to internal and external flux. 14
- 1(b). Determine the electric field of an infinite line of charge. 11
- 1(c). Determine the inductance of a 3- ϕ line operating at 50 Hz when the conductors are arranged as follows: 05



- 2(a). Why are suspension insulators preferred for high voltage power transmission? How guard ring method can be used to equalize the potential distribution among a string of suspension type insulator? 04+08=12
- 2(b). A 3- ϕ , 50 Hz overhead transmission line, 100 km long 110 kV between the lines at the receiving end has the following constants: 10
- Resistance per km per phase = 0.153 Ω , Inductance per phase = 1.21mH, Capacitance per km per phase = 0.00958 μF .
- The line supplies a load of 20,000 kW at 0.9 Power factor lagging. Calculate using nominal π representation, the sending end voltage, current, power factor, regulation and efficiency of the line. Neglect leakage.
- 2(c). A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, Calculate i) the distribution of voltage on the insulator discs and ii) The string efficiency. 08
- 3(a). What is meant by corona and critical disruptive voltage? How corona can be minimized in a transmission line? 10
- 3(b). What is sag in overhead lines? Discuss the disadvantages of providing too small or too large sag on a line. Why dampers are used in transmission line? 10
- 3(c). A transmission line over a hillside where the gradient is 1:20, is supported by two 22m high towers with a distance of 300m between them. The lowest conductor is fixed 2m below the top each tower. Find the clearance of the conductor from the ground. Given that conductors weighs 1kg/m and the allowable tension 1500kg. 10

SECTION – B

- 4(a). What are the various methods of voltage control in power system? Describe the off-load tap changing transformer method of voltage control and what are the limitations of the method? 03+7=10
- 4(b). Why are circuit breakers designed to have a short-time rating? Explain the principle of operation of a Vacuum Circuit Breakers (VCB). 14
- 4(c). A Circuit breaker is rated as 1500 A, 1000 MVA, 33 kV, 3 Second, 3 phase oil circuit breaker. Find i) rated normal current, ii) breaking capacity, iii) rated symmetrical breaking current, iv) rated making current, v) short-time rating. 06
- 5(a). What is a voltage surge? Discuss the causes of over voltages. 2+6=8
- 5(b). Describe the method of transformer grounding. 06
- 5(c). What is meant by sequence networks? Derive an expression for fault current for line-to-line fault by symmetrical components method and draw the sequence network connection for line-to-line fault. 3+10+3=16
- 6(a). The 3-phase power and line-line ratings of the electric power system shown in figure; 14



G_1:	60 MVA	20 kV	$X = 9\%$
T_1:	50 MVA	20/200 kV	$X = 10\%$
T_2:	50 MVA	200/20 kV	$X = 10\%$
M:	43.2 MVA	18 kV	$X = 8\%$
Line:		200 kV	$Z = 120 + j200\Omega$

Draw an impedance diagram showing all impedances in per-unit on a 100 MVA base. Choose 20 kV as the voltage base for generator.

- 6(b). What is the reason of transmission interconnection? What is the impact of cyber security on the smart grid? 4+4=08
- 6(c). Why is 3-phase symmetrical fault more severe than 3-phase unsymmetrical fault? 08

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2020

ESE 3123

(Thermo-Fluid Devices)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

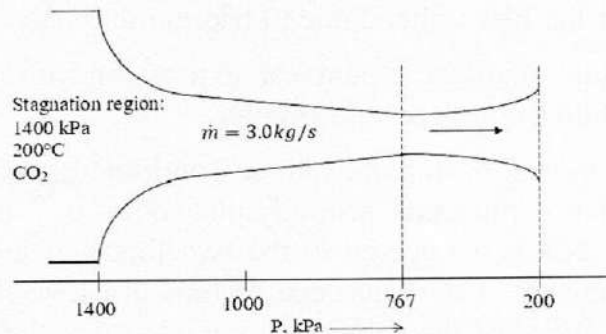
SECTION – A

- 1(a). What do you understand by laminar flow and turbulent flow? Derive an expression for the loss of head due to friction in pipes. 12
- 1(b). Define hydraulic diameter. Deduce an expression for measuring the actual flow rate of fluid through a venturimeter. 12
- 1(c). A submarine moves horizontally in sea and has its axis 15 m below the surface of water. A pitot tube properly placed just in front of the submarine and along its axis is connected to the two limbs of a U-tube manometer containing mercury. The difference of mercury level is found to be 200 mm. Find the speed of the submarine knowing that the specific gravity of mercury is 13.6 and that of sea water is 1.026 with respect of fresh water. 06
- 2(a). What is the reason of negative slip in reciprocating pump? 06
- 2(b). A single acting reciprocating pump has a plunger of diameter 250 mm and stroke of 350 mm. If the speed of the pump is 60 rpm and it delivers 16.5 liters per second of water against a suction head of 5 m and a delivery head of 20 m, find the theoretical discharge, coefficient of discharge, the slip, the percentage slip of the pump and the power required to drive the pump. 12
- 2(c). What will happen if the impeller of a centrifugal pump is made to rotate in the presence of air? 12
- 3(a). Write down the differences between fans, compressors and blowers. Also draw the main and operating characteristics curves of a centrifugal pump. 07
- 3(b). Define specific speed of a centrifugal pump. Deduce the expression for determining the specific speed of a centrifugal pump. 10
- 3(c). What do you understand by NPSH? Write down its significance. 05
- 3(d). Find the number of pumps required to take water from a deep well under a total head of 100 m. All the pumps are identical and are running at 900 rpm. The specific speed of each pump is given as 30 while discharge is $0.20 \text{ m}^3/\text{s}$. 08

SECTION – B

- 4(a). Prove that the velocity and flow area change in same direction for supersonic isentropic flow. 12
- 4(b). Differentiate between compressible and incompressible flow. 06
- 4(c). Carbon dioxide (CO₂) flows steadily through a varying cross-sectional area duct such as a nozzle shown in figure at a mass flow rate of 3.00 kg/s. The carbon dioxide enters the duct at a pressure of 1400 kPa and 200°C with a low velocity, and it expands in the nozzle to an exit pressure of 200 kPa. The duct is designed so that the flow can be approximated as isentropic. Determine the density, velocity, flow area and Mach number at the location of the duct where the fluid pressure is 1200 kPa. 12

For CO₂: $c_p = 0.846 \text{ kJ/kg.K}$, $k = 1.289$, $R = 0.1889 \text{ kJ/kg.K}$



- 5(a). What do you understand by Fouling factor? Briefly discuss about it. 06
- 5(b). Air is flowing through a duct and a normal shock wave is formed at a certain cross-section. The upstream Mach number, pressure and temperature are 2.2, 106 bar and 16°C, respectively. Calculate the Mach number, pressure and temperature immediately downstream of the shock wave. Take $k = 1.4$. 12
- 5(c). A counter flow double tube heat exchanger is used to heat water from 20°C to 80°C at a mass flow rate of 1.2 kg/s. The heating is to be accomplished by geothermal water available at 160°C at a mass flow rate of 2 kg/s. The inner tube is thin-walled and has a diameter of 1.5 cm. If the overall heat transfer coefficient of the heat exchanger is 640 W/m².K, determine the length of the heat exchanger required to achieve the desired heating. 12
- 6(a). Show that the general expression of effectiveness for parallel flow heat exchanger is given by: 20

$$\varepsilon = \frac{1 - e^{-\frac{U_o A_o}{C_{min}} \left(1 + \frac{C_{min}}{C_{max}}\right)}}{1 + \frac{C_{min}}{C_{max}}}$$

The symbols have their usual meaning.

- 6(b). A counter flow shell-and-tube heat exchanger is used to heat water at a flow rate of $\dot{m} = 0.8 \text{ kg/s}$ from $T_i = 30^\circ\text{C}$ to $T_o = 80^\circ\text{C}$, with hot oil entering at 120°C and leaving at 85°C . The overall heat transfer coefficient $U = 125 \text{ W/(m}^2\cdot^\circ\text{C)}$. Calculate the heat transfer area required. 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 3rd Year 1st Term Examination, 2020

ESE 3105

(Heat Transfer and Mass Transfer)

Time: 1 Hour 30 Minutes.

Full Marks: 120

N.B. i) Answer any TWO questions from each section.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

- 1(a). The inner surface at r_1 and the outer surface at r_2 of a hollow sphere are maintained at uniform temperatures T_1 and T_2 , respectively. The thermal conductivity of the solid is constant. Develop an expression for the radial heat flow rate through the hollow sphere. 15
- 1(b). A steel rod of diameter 2 cm, length 25 cm, and thermal conductivity $50 \text{ W/m}^\circ\text{C}$ is exposed to ambient air at $T_\infty = 20^\circ\text{C}$ with a heat transfer coefficient $64 \text{ W/m}^2\text{C}$. If one end of the rod is maintained at a temperature of 120°C , calculate the heat loss from the rod. 15
- 2(a). Define critical radius of insulation. Consider a pipe insulated in such a way that the outer radius of the insulation is less than the critical radius. Now the insulation is taken off. Will the rate of heat transfer from the pipe increase or decrease for the same pipe surface temperature. Justify your answer. 10
- 2(b). Explain why the condenser tubes are usually horizontal. 05
- 2(c). Suppose a cylindrical aluminium fin with length L and thermal conductivity k exposed to a medium having temperature T_∞ with a heat transfer coefficient h . If the heat loss from the fin tip is negligible and fin base temperature is T_b , derive an expression for the efficiency of the fin. 15
- 3(a). Air free saturated steam at 85°C and 57.83 kPa condenses on the outer surface of 225 horizontal tubes of 1.27 cm OD arranged in a 15 by 15 array. Tube surfaces are maintained at a uniform temperature of 75°C . Calculate the total condensation rate per meter length of the tube bundle. 15
- [Given, the physical properties of water at 80°C are: thermal conductivity $= 0.668 \text{ W/m}^\circ\text{C}$, dynamic viscosity $= 0.355 \times 10^{-3} \text{ kg/m.s}$, density $= 974 \text{ kg/m}^3$, latent heat of vaporization $= 2309 \text{ kJ/kg}$]
- 3(b). Draw the boiling curve and identify different boiling regimes. Also identify the burn-out point on the curve. Explain how burn-out is caused. Why is the burn-out point avoided in the design of boilers? 15

SECTION – B

- 4(a). What is the physical significance of Nusselt number and Prandtl number? 10
- 4(b). Consider a laminar flow over a flat plate with the following boundary conditions: 20

$$u(0, y) = V; u(x, 0) = 0; v(x, 0) = 0; u(x, \infty) = V;$$

$$T(0, y) = T_{\infty}; T(x, 0) = T_w; T(x, \infty) = T_{\infty}$$

Determine the velocity boundary layer thickness.

- 5(a). For a laminar flow over a flat plate, develop an expression to determine the average friction coefficient. 10
- 5(b). The forming section of a plastic plant puts out a continuous sheet of plastic that is 1.2 m wide and 2 mm thick at a rate of 15 m/min. When the plastic sheet is exposed to the surrounding air, the temperature of sheet is 90°C and the sheet is subjected to air flow at a velocity of 3 m/s at 30°C on both sides along its surfaces normal to the direction of the sheet. The width of the air-cooling section is such that a fixed point on the plastic sheet passes through the section in 2 second. Determine the rate of heat transfer from the plastic sheet to the air. Following data can be used: 20

For plastic sheet: $\rho = 1200 \text{ kg/m}^3$, $c_p = 1.7 \text{ kJ/kg} \cdot ^\circ\text{C}$, $\varepsilon = 0.9$

For air:

Temp. ($^\circ\text{C}$)	$k \text{ (W/m} \cdot ^\circ\text{C)}$	$\nu \text{ (m}^2/\text{s)}$	Pr	$c_p \text{ (J/kg} \cdot ^\circ\text{C)}$
30	0.02588	1.608×10^{-5}	0.7282	1007
40	0.02662	1.702×10^{-5}	0.7255	1007
80	0.02953	2.097×10^{-5}	0.7154	1008
90	0.03024	2.201×10^{-5}	0.7132	1008

- 6(a). Define mass diffusivity. Explain Fick's law of diffusion. Also discuss some common analogies between heat and mass transfer. 12
- 6(b). What is the physical significance of Grashof number? How does it differ from Reynolds number? Explain. 08
- 6(c). Is there any difference between mass transfer and bulk fluid flow? Is it possible to occur mass transfer in a homogeneous medium? Explain your answer. 10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

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Math 3113

(Statistics and Numerical Methods)

Time: 1 Hour 30 Minutes.

Full Marks: 120

- N.B. i) Answer any TWO questions from each section.
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SECTION – A

- 1(a). Discuss round-off errors and truncation errors. 05
- 1(b). Find the root of the equation $x^2 - 3x + 2 = 0$ by using Newton-Rapson method with initial guess $x = 0$ up to six iteration. 10
- 1(c). Obtain the solution of the following system of linear equation using Gause-Seidel iteration method (show only two iteration) 15

$$2x + y + z = 5; 3x + 5y + 2z = 15, 2x + y + 4z = 8$$

- 2(a). The table below gives the temperature T ($^{\circ}\text{C}$) and lengths l (mm) of of heated rod. Fit a straight line for the best values of the coefficients regarding T as independent variables: 17

T	30°	40°	50°	60°	70°	80°
l	800.3	800.4	800.6	800.7	800.9	802.0

- 2(b). Determine a root of the equation $x^2 - 4x - 10 = 0$ using bisection method. 13
- 3(a). Use Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rules to evaluate $\int_0^{\pi/2} \sqrt{\sin(x)} dx$ 12
- 3(b). Using classical Runge-kutta method to estimate $y(0.4)$ 18
when $y'(x) = x^2 + y^2$; $y(0) = 0$; assuming $h = 0.2$

SECTION – B

4. Find first moments measured from origin. Hence find mean, standard deviation, coefficient of variation, coefficient of skewness and coefficient of kurtosis of the following data: 30

3, R, 1, R;

Where R is the sum of last two digits of your class roll number [Hints: Roll 1713024, then $R = 2 + 4 = 6$]

- 5(a). Test whether the following functions are probability function or not for same value of R. 10

X	-1	0	1	2
P(x)	2R	R	2R	0.5

Hence find $P(x \geq 1)$. R is the sum of last two digits of your class roll number.

- 5(b). If possible, find (i) $E(5x^2 + 2)$ and (ii) Standard deviation of the problem given below: 20

$$g(x) = \begin{cases} \frac{1}{(b-R)} & \text{if } R \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Where b is sum of all digits of your class roll and R is the sum of last two digits of your class roll number.

[Hints: Roll 1713024, then $b = 1+7+1+3+0+2+4 = 18$, and $R = 2 + 4 = 6$]

- 6(a). What is meant by the Binomial distribution? If the mean and standard deviation of a Binomial distribution are 40 and 6 respectively, then calculate n, p and q. 15
- 6(b). What are the conditions that the Binomial distribution tends to Poisson distribution? 03
- 6(c). The number of arrivals of customers during any day follows Poisson distribution with a mean of 5. What is the probability that the total number of customers on two days selected at random is less than 2? 12

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